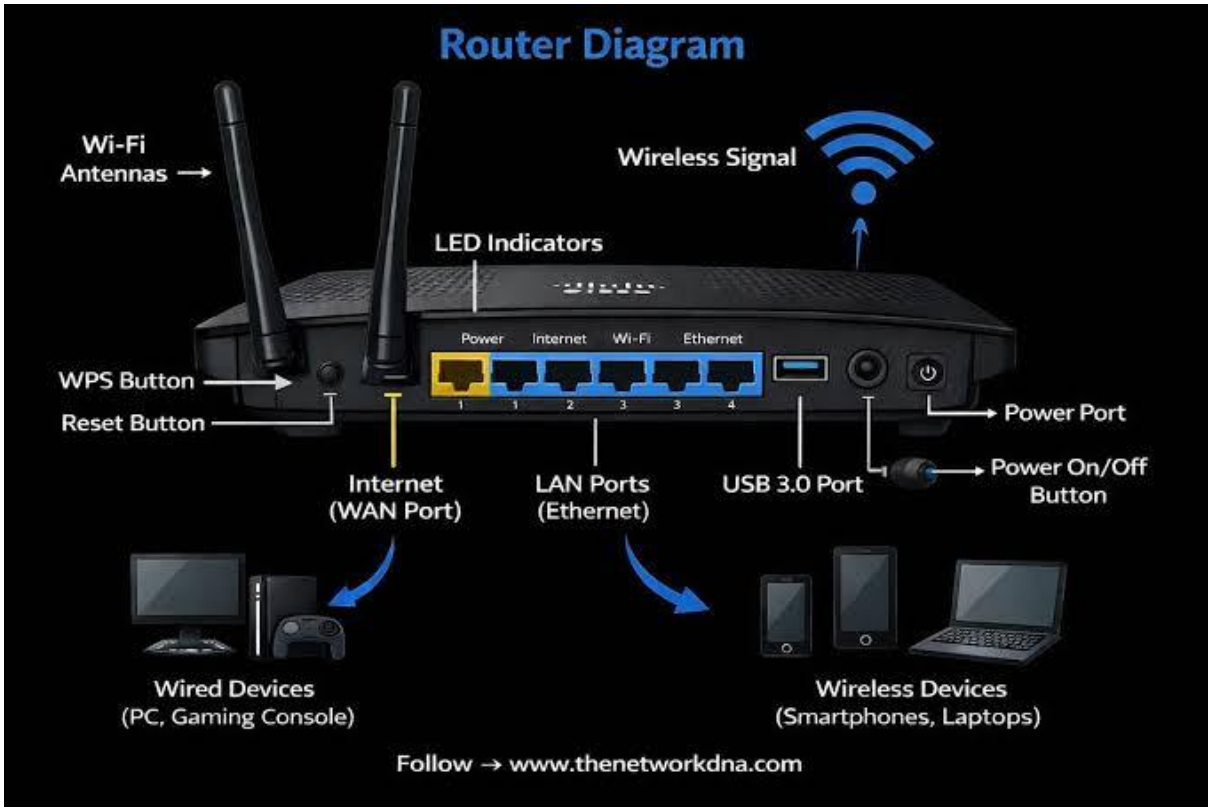
Assignment 9

1.Draw a simple network diagram showing how your smartphone connects to Instagram — include your device, Wi-Fi router, ISP, and Instagram's server, and label each part.

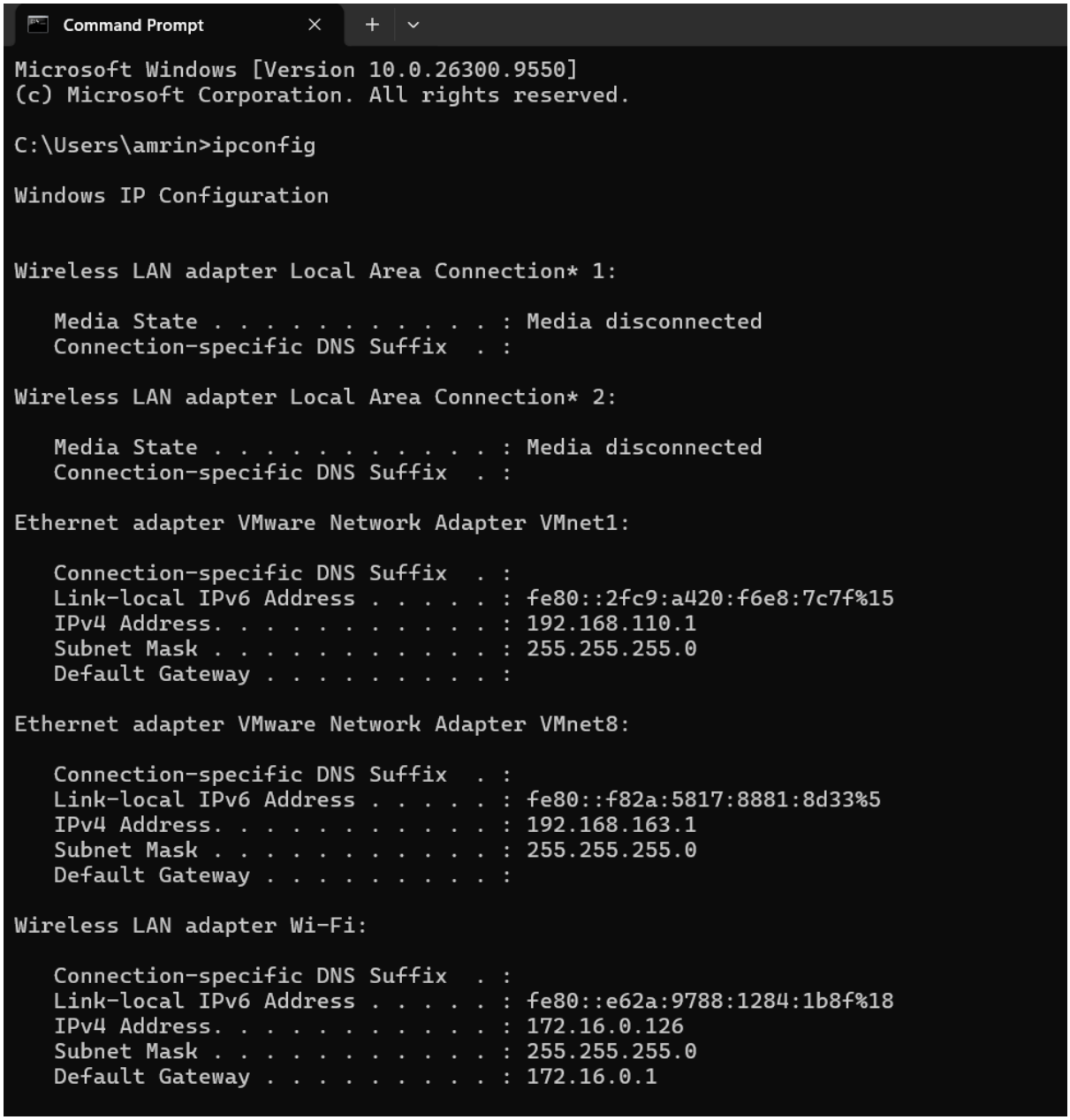
Ans-



Part	Function
Smartphone	Runs the Instagram application
Wi-Fi Router	Connects your device to the local network and Internet
ISP	Provides Internet connectivity
Internet	Carries data between networks
Instagram Server	Processes Instagram requests and sends data back

2.List the IP address of your device and your default gateway using the 'ipconfig' command (Windows) or 'ifconfig' (Linux/Mac), and explain what each means in 1-2 lines.

Ans-



**3.Simulate a Zomato food order journey by writing the steps/data packets that travel from your device to Zomato's server and back, mentioning protocols like HTTP, TCP/IP, and DNS wherever relevant.

Hint: Think about what happens when you search for a restaurant and place an order.**

Ans-

Step 1: Open Zomato

Open the Zomato application or website.

For example, you search:

Pizza

Your device creates a request to the Zomato service.

Step 2: DNS Resolution

Your device needs to find the IP address associated with Zomato's domain.

DNS means:

Domain Name System

It converts a domain name into an IP address.

Step 3: TCP Connection

After obtaining the server's IP address, your device establishes a network connection.

TCP provides:

- Reliable communication
- Error checking
- Ordered delivery
- Connection management

For secure web traffic, HTTPS is normally used rather than plain HTTP.

Device → TCP connection → Zomato Server

Step 4: Search Restaurant

You search for:

Pizza near me

Step 5: Zomato Sends Restaurant Data

The server sends information back, such as:

Restaurant Name

Menu

Price

Rating

Delivery information

Step 6: Select Food

You select a restaurant and add food to your cart.

Step 7: Place Order

You click: Place Order

Protocol	Purpose
DNS	Converts domain name to IP address
IP	Provides addressing and routing
TCP	Reliable transport of data
HTTPS	Secure web communication
HTTP	Web application communication; HTTPS is HTTP over TLS

4.Use the 'ping' command to check the connection between your computer and www.spotify.com, then explain what the results mean for network troubleshooting.

Ans-

